A Multi Strategy Approach to Trading Foreign Exchange Futures

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January 25, 2019

Abstract

In this article we present a systematic multi-strategy approach to trading foreign exchange futures for a managed futures portfolio. Our central finding is that there is more alpha to be derived from combining different indicators compared to hand engineering each indicator. We show that combining technical indicators like momentum and mean reversion with fx carry indicators leads to significant improvement over individual indicators. Through an end to end systematic portfolio construction methodology, including indicator construction, normalization and combination we are able to improve the Sharpe Ratio of the resulting portfolio over the best performing single indicator by 60% when evaluated in an unbiased walk forward backtest.

Foreign Exchange Futures Portfolio Construction Managed Futures

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1 Introduction

Our belief is that the optimal strategy for trading foreign exchange futures is a combination of technical and macro/fundamental strategies. This belief is backed by the strong research validating a wide array of risk factors affecting the returns of foreign exchange futures. The carry trade or the strategy of borrowing money from a low interest rate economy and lending it to a high interest rate economy has been used by traders and dealers for ages. This has also been verified by research, most notably by Fama [Fam84]. The asset pricing anomaly called *time series momentum* has also been validated to be consistent across asset classes, Moskovitz et al [MOP12] highlight the persistence in returns for 1 to 12 months that partially reverses over longer horizons, consistent with sentiment theories of initial under-reaction and delayed over-reaction. In the global economy, asset managers borrow and sell currencies to trade international equities, the relationship between equity market returns and foreign exchange returns is proven by Karoui Aymen [Aym06]. Commodity prices, specifically agricultural commodities, oil and metal effect foreign exchange returns significantly, Joseph P Byrne et. al. [BIS17] highlight this anomaly looking at the developed and emerging markets.

We evaluate the carry anomaly[1], momentum and mean reversion in foreign exchange future prices as well as momentum of linked equity indices and commodities to build a multi-strategy portfolio. We normalize and convert these indicators to returns while scaling the risk to a uniform level and then combine these indicators using various combination strategies like equal weighting, maximizing diversification, optimization for maximizing Sharpe ratio etc. We evaluate the returns of the final strategy in an unbiased walk-forward fashion.

While our research is an end-to-end systematic way of constructing a foreign exchange futures trading strategy, the underlying theme of the strategy comes from carry anomaly widely used by discretionary traders for foreign exchange trades combined with conviction that behavioral indicators like momentum of the foreign exchange futures and linked equity and commodity indices come to play in the short and the long term.

This strategy is built to be traded as a part of a managed futures portfolio. Managed futures strategies are popular multi-asset derivative strategies which due to their low correlation to the traditional equity portfolios enhance returns on a risk adjusted basis and reduce the overall portfolio volatility. [SSS12]

2 Related Literature

The multi strategy approach to trading FX futures is relatively new but there are some notable research based approaches published to capture foreign exchange futures market risk premia which deserve a mention.

Serban, 2009 [Ser10] explores a combination of mean reversion and momentum strategies for trading foreign exchange futures. Building on the research by Fama and French, 1988 [FF88] and Summers, 1986 [Sum86] in the equity market, they establish the evidence of momentum and mean reversion effects in deviations of from Uncovered Interest Parity and show that a combination strategy using momentum and mean reversion outperforms a pure momentum and a pure mean reversion strategy.

Ahmerkamp et al [AG13] provide evidence that speculative capital of hedge funds is a key determinant for the profitability of optimal carry and momentum strategies in futures markets across asset classes. They construct optimal carry and momentum portfolios and allocate to them from the perspective of a utility maximizing risk averse

[1]Carry anomaly refers to excess returns from the carry trade i.e. by borrowing in low interest currencies and lending high interest rate currencies

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investor. They use a parametric approach in constructing their optimal portfolios and demonstrate that their approach enhances the portfolio Sharpe of FX futures significantly.

Olszweski et al [OZ13], look at a combination of momentum and carry from a managed futures portfolio perspective as well. They evaluate the source of returns for both these strategies, their correlations and the optimal combination methodology for constructing a diversified portfolio.

Barosso et all [BSC15] test the relevance of technical and fundamental variables in forming currency portfolios. They demonstrate that carry, momentum and reversal all contribute to portfolio performance, whereas the real exchange rate and the current account do not. They also show that exposure to currencies increases the Sharpe ratio of diversified portfolios by 0.5 on average, while reducing crash risk.

3 Methodology

The methodology used is as summarized in Fig. 1. In the following sections, we describe of each of these steps in more detail.

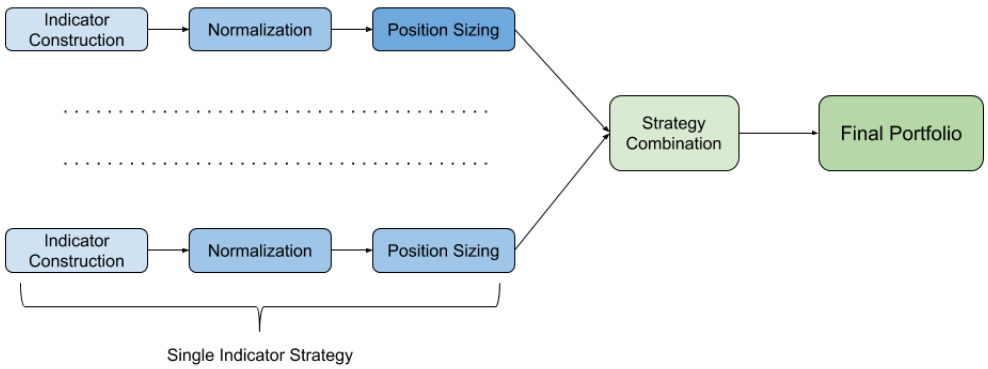


Figure 1: Portfolio Construction Methodology : First we create strategies that are based on individual indicators through indicator construction, normalization, and position sizing. Then we combine individual strategies together to get the final portfolio.

3.1 Instrument Selection

We use the 8 most liquid current month FX futures on the Chicago Mercentile Exchange (CME) for this analysis. The data is obtained from the CME website and we apply a T-3 day rollover to obtain a uniform series. We use daily OHLC data since the beginning of 1995.

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Table 1: FX Future Instruments Used

Currency	Explanation	Linked Equity Index
6A series	Australian Dollar Futures	ASX 200
6B series	British Pound Futures	FTSE 100
6C series	Canadian Dollar Futures	TSX Composite Index
6E series	Euro FX Futures	Euro STOXX 50
6J series	Japanese Yen Futures	Nikkei 225
6M series	Mexican Peso Futures	SP/BMV IPC
6N series	New Zealand Dollar Futures	NZS 50 Index
6S series	Swiss Franc Futures	Swiss Market index

3.2 Indicator Selection and Construction

The first step is to select indicators that will be used to create candidate strategies for combination at a later stage. Based on literature review and our research, we found the following indicators to be useful for modelling currency futures returns. We create two indicators - short term and long term, for each of the following risk premia.

3.2.1 Interest Rate Carry

The carry anomaly driving foreign exchange future returns can be explained due to the interest rate arbitrage opportunity present where a trader can borrow at low interest rate and lend at a higher interest rate in different geographies, which drives the currency prices [Fam84] We use the log difference of the long term (10 years) and short term (1 year) yields between the two geographies to get the carry indicator.

These indicators are referred to as *short term yield difference* and *long term yield difference* below.

3.2.2 Momentum

The momentum anomaly is validated across asset classes with ample research. It is the empirically observed tendency for rising asset prices to rise further, and falling prices to keep falling. We use a short term as well as long term momentum indicator modelled as 3 month log returns and 12 month log returns respectively.

These indicators are referred to as *short term momentum* and *long term momentum* below.

3.2.3 Mean Reversion

The cross-currency rates between developed currencies tend to mean revert to a particular level. We use the returns with respect to a short term moving average (3 months) and a long term moving average (12 months) to model short-term and long-term mean reversion.

These indicators are referred to as *short term mean reversion* and *long term mean reversion* below.

3.2.4 Equity Momentum

We look at short term (3 months) and long term (12 months) momentum as predictive variables governing foreign exchange rates. Commodity currencies and emerging market currencies are known to move with major risk on/risk off moves in the equity markets. Karoui et.al. [Aym06] also provide evidence of correlation between foreign exchange rate volatility and stock exchange returns volatility.

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These indicators are referred to as *short term equity momentum* and *long term equity momentum* below.

3.2.5 Commodity Momentum

Byrne te al [BIS17] investigate the importance of commodity prices to the returns of currency carry trade portfolios. They show that agricultural material and metal price risk factors are found to have explanatory power on the cross-section of currency returns, while commodity common and oil factors do not. We look at SP Goldman Sachs Commodity Index (GSCI) as a common commodity index along with Brent crude, gold and agriculture index rates momentum in the short term (3 months) and long term (12 months).

These indicators are referred to as *short term commodity momentum* and *long term commodity momentum* below.

In addition to the above mentioned we add short term (3 months) and long term (12 months) indicators based on realized volatility of each security's returns.

3.3 Normalization

Different indicators as described above are in different units. In order to use a common position sizing methodology for all indicators, we normalize the raw indicator values so that they all lie in the range $[-0.5, 0.5]$. To achieve this, we do a walk forward percentile normalization across time to the indicators followed by. In other words, we calculate the percentile score of the indicator on a given day with respect to all previous indicator values which brings the indicator to a range $[0, 1]$. This is followed by subtracting 0.5 to remove any long bias in the strategy over time. The transformation can be written as follows.

$$NI_s^t = Percentile(I_s^0 ... I_s^{t-1}) - 0.5$$

where I_s^0 is the indicator value for security s at the beginning which refers to the time since we have data
I_s^t is the indicator value for security s at time t,
NI_s^t is the normalized indicator value for security s at time t,
and this is done daily so increments in t are in days.

3.4 Position Sizing

Once we have the normalized indicator values in the range $(-0.5, 0.5)$, we use risk budgeting to come up with allocation for each indicator. The risk budgets are taken to be proportional to the absolute value of normalized indicator values and total risk of the allocation is set to 10% annualized for all indicators. A constant risk target is used across all indicators to ensure that returns from each indicator are comparable when we allocate across different indicators in the following section.

$$RB_s^t = (|NI_s^t| / (\sum_s |NI_s^t|)) * \tau$$

where, τ is the target risk of the allocation (10%)
RB_s^t is the risk budget for security s at time t

However, as opposed to the common practice of having positive allocation for all securities in risk budgeting allocation, we retain the sign of the normalized indicator

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values and allow negative allocation. We solve the following modification of risk budgeting optimization to allow for negative weights. Dropping the index t over time below for ease of notation.

$$\begin{gathered}\text{Find weights } w_s \text{ that minimize} \\ \textstyle\sum_s (RC_s - RB_s)^2 \\ RC_s = w_s * (\textstyle\sum \cdot w)_s \\ \text{such that, } w^T \cdot \textstyle\sum \cdot w = \tau, \\ w_s * sign(NI_s) \geq 0\end{gathered}$$

where, RC_s is the risk contribution due to security s
$\sum$ is the covariance matrix over the traded securities[2].

In practice, leverage constraints are added as part of the optimization.

3.5 Strategy Combination

This section deals with the major contribution of this paper - how to combine different indicators together. We already have strategies based on individual indicators that are scaled to uniform risk. In this step, we discuss different ways these strategies can be combined in a systematic manner to outperform individual strategies.

Individual indicators are not orthogonal, so finding the optimal combination using a multiple linear regression model or optimization could be problematic. The correlation between the return series based on the indicators is shown in Table 2

Indicator	Long term yield diff	Short term yield diff	Short term momentum	Long term momentum	Short term mean reversion	Long term mean reversion	Short term volatility	Long term volatility	Short term equity momentum	Long term equity momentum	Short term commodity momentum	Long term commodity momentum
Long term yield diff	100.00	68.54	26.88	42.13	22.76	38.92	24.07	24.72	-8.30	-19.12	11.29	21.00
Short term yield diff	68.54	100.00	20.77	36.01	16.10	31.76	28.35	24.84	-16.17	-23.50	-1.00	7.56
Short term momentum	26.88	20.77	100.00	56.35	83.34	76.26	-1.88	1.03	25.24	8.92	38.12	18.04
Long term momentum	42.13	36.01	56.35	100.00	49.86	82.91	-8.02	-6.75	13.21	20.42	32.22	42.61
Short term mean reversion	22.76	16.10	83.34	49.86	100.00	70.96	-1.57	0.77	26.06	9.40	38.73	17.45
Long term mean reversion	38.92	31.76	76.26	82.91	70.96	100.00	-5.50	-2.34	22.18	17.48	37.83	31.44
Short term volatility	24.07	28.35	-1.88	-8.02	-1.57	-5.50	100.00	69.55	-10.38	-27.07	1.23	-6.25
Long term volatility	24.72	24.84	1.03	-6.75	0.77	-2.34	69.55	100.00	-7.47	-21.71	2.40	-2.59
Short term equity momentum	-8.30	-16.17	25.24	13.21	26.06	22.18	-10.38	-7.47	100.00	51.43	27.29	6.74
Long term equity momentum	-19.12	-23.50	8.92	20.42	9.40	17.48	-27.07	-21.71	51.43	100.00	18.43	24.77
Short term commodity momentum	11.29	-1.00	38.12	32.22	38.73	37.83	1.23	2.40	27.29	18.43	100.00	54.99
Long term commodity momentum	21.00	7.56	18.04	42.61	17.45	31.44	-6.25	-2.59	6.74	24.77	54.99	100.00

Table 2: The above table shows cross-correlation of different indicators. The correlation is computed on the returns of the respective single indicator based strategies constructed using the process highlighted in Fig. 1

To avoid multi-collinearity we avoid simple regression based prediction framework and look at alternate methods for indicator combination which improve risk adjusted performance in out of sample data. In order to combine different strategies, we do a monthly rebalancing of the portfolio looking at the past data using the following methods.

3.5.1 Equal weighting

Equal weight given to all the strategies. Even though this method is fairly simple, it performs well in many applications. However, in an systematic setting where the list of input strategies are not curated in terms of performance using equal weighting is likely to have worse performance. Another shortcoming of equal weighting scheme is that it ignores the correlation across different strategies which could lead to betting heavily on similar risk factors and lower diversification.

[2]How we computed the covariance matrix can be found in the Appendex

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3.5.2 Risk parity

Strategies are given weights proportional to the reciprocal of the realized volatility of the return series. As the strategies have already been scaled to uniform target risk, empirically it could be quite similar to equal weighting. This method suffers from similar issues as equal weighting method.

3.5.3 Equal risk contribution

In this method, strategies are assigned weights such that each strategy has equal risk contribution. This method is a special case of the risk budgeting position sizing described earlier in the previous section. This method looks at the covariance matrix and therefore is more diversified as compared to risk parity and equal weighting.

3.5.4 Proportional to Sharpe

Strategies are assigned weight proportional to their realized Sharpe till that point in time. Although this method is robust against presence of underperforming strategies, it overlooks the correlation across strategies and therefore can lead to concentration on certain risk factors.

3.5.5 Correlation aware Sharpe

Strategies are assigned weights proportional to their realized Sharpe till that point in time minus the sum of their correlation with all other indicators. This is a simplified method to rank based on performance while penalizing high correlation, thus being robust against under-performing strategies while avoiding concentration of certain risk factors.

3.5.6 Maximum diversification

We assign weights to maximize the diversification ratio of the portfolio. The diversification ratio is defined as the weighted average of the volatilities of the components of the portfolio divided by the portfolio volatility. This method tries to create a portfolio where the risk is as diversified as possible without taking into account the expected returns.

3.5.7 Volatility scaled negative correlation

The negative correlation or the inverse of the correlation matrix is a contrary relationship between two variables such that they move in opposite directions. The strategies are assigned weights proportion to the inverse of the correlation matrix divided by the volatility matrix. This method avoids concentration of risk factors while being aware of the volatility of the strategies.

3.5.8 Negative correlation dot Sharpe

This is a more mathematically robust version of Correlation aware Sharpe method described above. The strategies are assigned weights proportion to the inverse of the correlation matrix multiplied by the their realized Sharpe till that point. This method is robust against under-performing strategies while avoiding concentration of certain risk factors.

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3.5.9 Optimization to maximize Sharpe

In this method, we find the strategy weights by optimizing realized sharpe ratio of the combination on the data till that point in time. This method could be somewhat susceptible to overfitting since aggregate sharpe ratio could be greatly affected by extreme (positive or negative) returns during a certain period in the backtest. The next method tries to address this shortcoming.

3.5.10 Optimization to maximize 10th percentile of Sharpe

In this method, we take the daily returns series of each strategy and on each day we compute the sharpe ratio over last one year which gives us series of rolling one year sharpe ratio. We take 10th percentile of this series as our optimizing metric. The motivation is that optimizing this metric favors strategies that have consistently good returns as opposed optimizing an aggregate metric that can easily be influenced by outsized returns in a certain period in the backtest. Moreover, by favoring consistent returns, this method is less likely to overfit to noise in historical data.

4 Results

We first look at the returns of single indicator based strategies to evaluate the strengths of the indicators selected. It is important to realize these strategies were not optimized for performance. Table 3 summarizes the performance [3] of different single indicator strategies.

Next, we look at the results for combination of multiple single indicator strategies. We evaluate these combination strategies in two ways. First by getting an indicator value per product which is a weighted combination of single indicator values, keeping same weights across products. This is done with the purpose of minimizing the degrees of freedom so as to reduce any overfitting that may arise from idiosyncratic movements in one currency pair. In the second method, we increase the degrees of freedom and combine a larger set of return series of indicator and product combinations.

If we were to find optimal combination of strategies in in-sample data the combination strategies would trivially out-perform single indicator based strategies and the combination strategies with higher degrees of freedom would perform even better. However, given that we are evaluating all the methods in a walk-forward setting, even matching the returns of the best strategy would be equivalent to finding the best expert.

4.1 Single indicator strategies

The returns for the strategies based on a single indicator can be seen in Table 3.

4.2 Combination strategies with uniform weights across products

Using the combination methods listed in Section 3.5, we compute weights per indicator (but enforcing same weights across products). The results of the various combination methods are summarized in Table 4 and Fig. 3 shows the plots of cumulative returns of different methods.

As shown in Table 4, maximizing 10th percentile of the one year rolling sharpe yields best performance and outperforms the best single indicator strategy by a margin. This

[3] Annualized sharpe ratio is computed by subtracting US 1-month interest rate to get excess returns series

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Indicator	Annualized Return	Annualized Sharpe
Long term yield diff	7.52	0.55
Short term commodity momentum	5.86	0.39
Short term yield diff	5.44	0.35
Long term momentum	4.42	0.25
Short term volatility	3.41	0.15
Long term commodity momentum	3.13	0.13
Short term equity momentum	3.06	0.12
Long term mean reversion	2.99	0.11
Short term momentum	2.18	0.03
Long term equity momentum	0.59	-0.12
Long term volatility	0.03	-0.18
Short term mean reversion	-0.23	-0.21

Table 3: Annualized returns and sharpe of strategies based on single indicators. Each strategy is scaled to have 10% annualized volatility. The allocations are obtained from indicator values as described in Section 3.

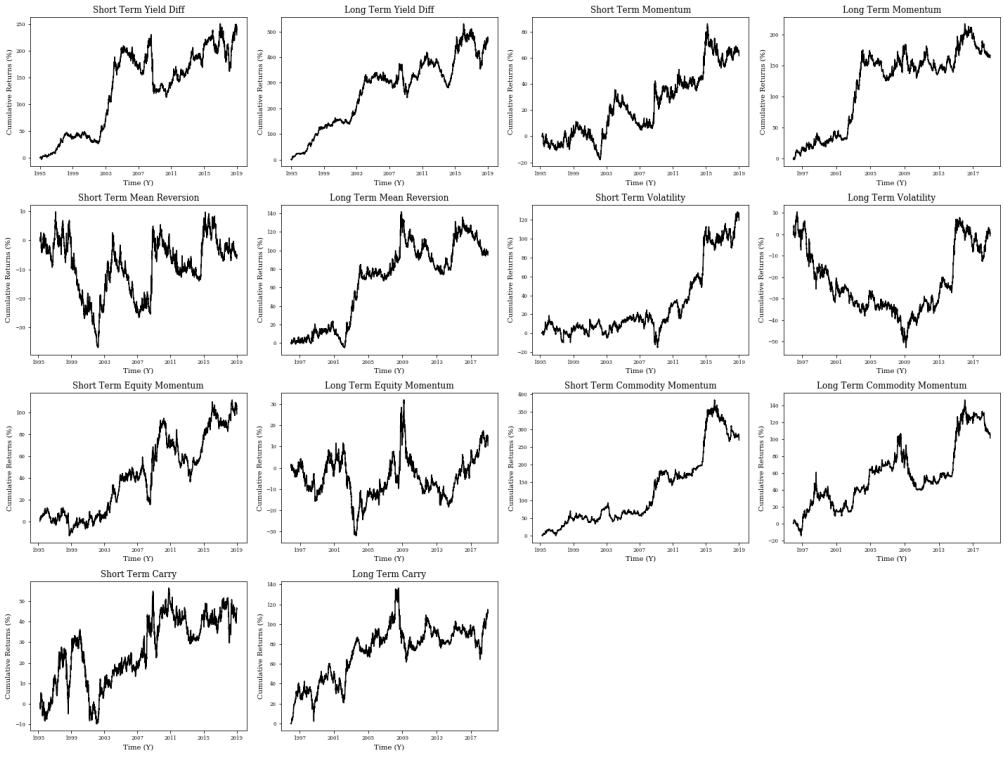


Figure 2: The above figure shows the cumulative returns of single indicator strategies. The long term yield difference shows the best returns followed by short term commodity momentum and short term yield difference. Moreover, the performance across different periods in the backtest is quite different which can be leveraged by combining these strategies.

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Combination Strategy	Annualized Return	Annualized Sharpe
Volatility scaled negative correlation	8.24	0.67
Max diversification	7.61	0.60
Equal risk	6.79	0.52
Equal risk contribution	6.57	0.50
Equal weight	6.58	0.50
Correlation Aware Sharpe	6.02	0.45
Proportional to Sharpe	5.91	0.44
Optimization to maximize 10th percentile of Sharpe	5.02	0.35
Negative correlation dot Sharpe	1.50	0.00
Optimization to maximize Sharpe	1.39	-0.01

Table 4: Performance of different methods for combining single indicator strategies with uniform weights across products. Methods which assign strategy weights by optimizing a conservative metric which enforces diversity across strategies perform better in out of sample data.

is quite a significant results considering the difficulty of the problem at hand. The combination method has no knowledge which strategy is could do well in future let alone know the best combination. Fig. 4 shows the weights assigned by this method to different strategies over time. Fig. 5 shows a snapshot of average weights and the latest weights assigned to indicators. The non-triviality of the result is clear when one looks at the dismal performance of the method that simply maximizes sharpe to assign weights to strategies.

A common pattern that emerges from the results in Table 3 is that methods which optimize a conservative metric which enforce diversity across strategies tend to outperform in out of sample period. It is important to understand that maximizing 10th percentile of rolling one year sharpe effectively looks at all the data points compared to looking at an aggregate metric like sharpe over the entire duration is hence able to achieve much better generalization in out of sample data.

4.3 Combination strategies on product, indicator combinations

Next we add more degrees of freedom to our approach to set one indicator and one product combination as one strategy, which would increase our strategy set to number of products x number of indicators. This is done to account for the fact that different currency pairs are affected by different risk factors such as commodity currencies (AUD, CAD, MXN, NZD) being affected commodity prices.

We heuristically remove candidate strategies from our strategy set that exhibit bad performance before we combine strategies. Ideally, we would like to examine each currency pair to identify relevant risk factors, but we use a performance based pruning as a proxy for removing irrelevant indicators. The set of strategies (product, indicator combinations) available for the strategy combination stage after pruning are listed in Table 5.

The performance of different combination methods are as summarized in Table 6 and Fig. 6. Here again we observe that maximizing diversification and maximizing the 10th percentile of Sharpe are the best performing methods for combining strategies. The sharpe ratio of the best combination portfolio is 0.3 or 60% higher than the best single indicator strategy. We observe that the ranking of different methods in terms of performance is similar to previous setting where same indicators are shared by all currency pairs.

As shown in Table 6 and Table 4, the performance improves significantly after allowing different currency pairs to use different indicators. In a walkforward evaluation, one would expect the increase in search space to adversely affect performance as borne out

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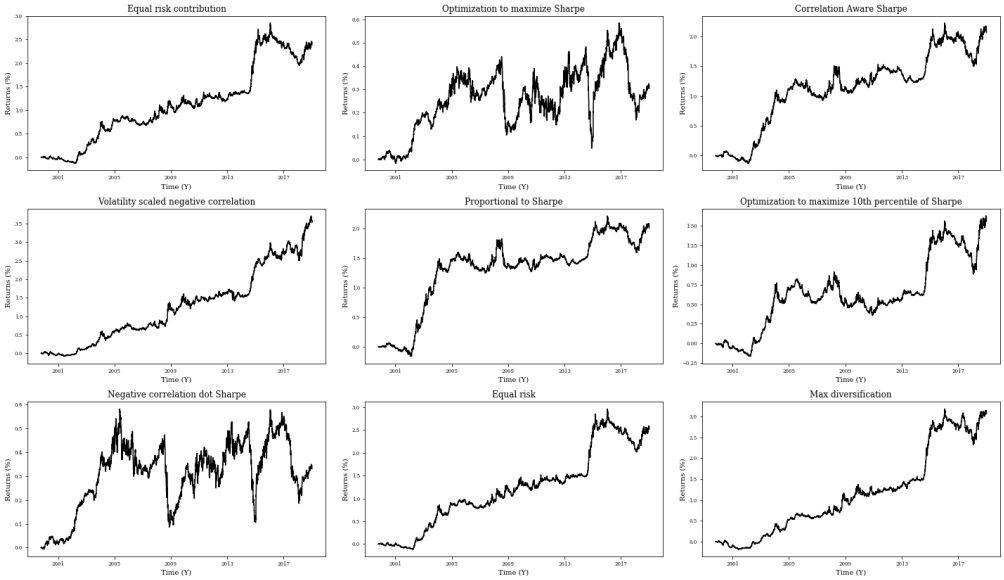


Figure 3: Cumulative returns of different methods of combining strategies when using same weights across products. The combination methodology which maximizes rolling 10th percentile of the one year sharpe shows the best returns, with Sharpe 0.1 higher than the best single indicator based strategy.

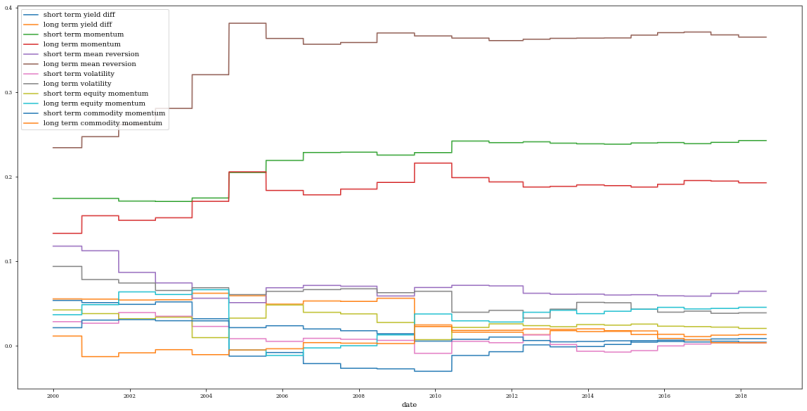


Figure 4: Weights assigned to different single indicator strategies by the method *Volatility scaled negative correlation*

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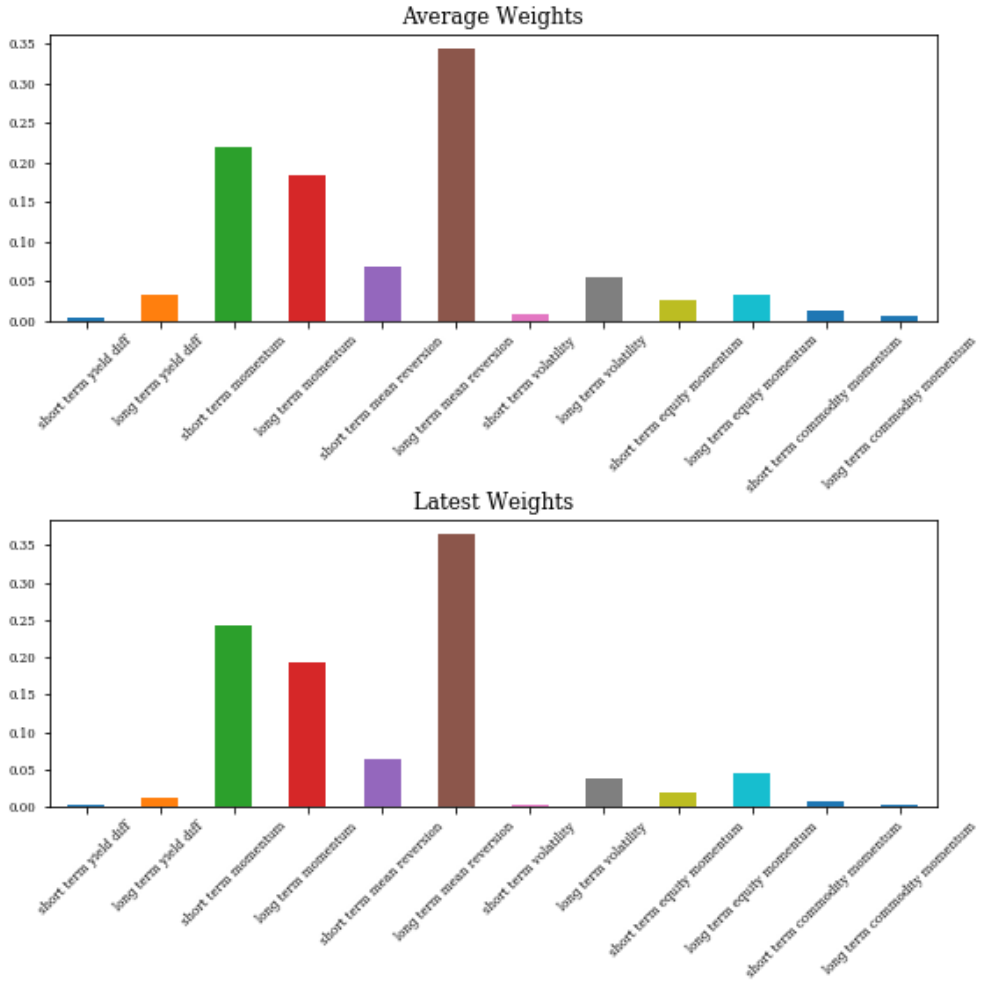


Figure 5: Average weights and latest weights assigned to different single indicator strategies by the method *Volatility scaled negative correlation*

Product	Indicators
6A series	long term mean reversion, long term momentum, long term yield diff
6B series	long term equity momentum, short term carry, short term equity momentum
6C series	short term equity momentum, short term volatility
6E series	long term yield diff
6J series	long term volatility, short term commodity momentum, short term momentum, short term volatility
6M series	short term yield diff
6N series	long term yield diff
6S series	long term yield diff, long term yield diff, short term commodity momentum, short term yield diff

Table 5: Indicators selected for different products based on performance based heuristics. It turns out that the long term yield diff is a good indicator for most of the currency pairs.

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	Annualized Returns	Annualized Sharpe
Volatility scaled negative correlation	10.73	0.88
Max diversification	10.33	0.84
Optimization to maximize 10th percentile of Sharpe	9.26	0.74
Equal risk contribution	9.19	0.73
Equal weight	9.19	0.73
Equal risk	8.39	0.65
Correlation Aware Sharpe	7.16	0.53
Proportional to Sharpe	6.79	0.49
Optimization to maximize Sharpe	6.15	0.43
Negative correlation dot Sharpe	2.04	0.03

Table 6: Annualized returns and sharpe of different methods for combining single indicator strategies with product x indicators as candidate strategies. The ranking of different methods is similar to the earlier case where weights were same across products.

by the method which optimizes sharpe ratio where sharpe ratio goes down from 0.11 to -0.21 but we find that a number of methods show significant performance improvement in this setting.

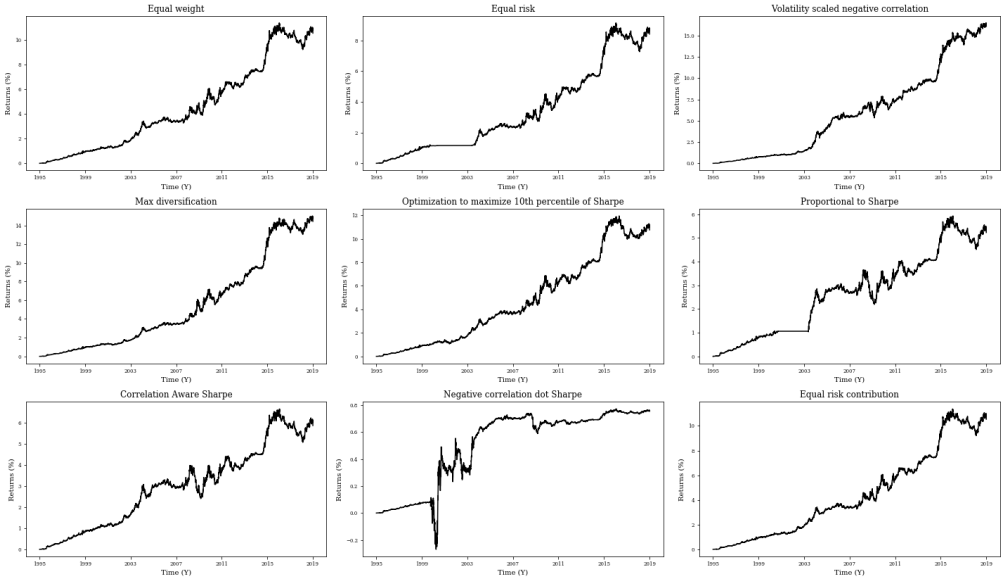


Figure 6: Cumulative returns of different methods of combining strategies when using same weights across products.

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5 Conclusion

In line with our thesis we find that a multi-strategy approach is a better approach than trading a single strategy as it leads to more consistent returns. Using our systematic approach we are able to enhance the sharpe ratio of the combination by up to 0.3 or 60% over the best performing strategy.

The indicator set we selected are significant in forecasting futures returns with the best single indicator strategy for trading foreign exchange futures being the long term carry in interest rates followed by short term commodity momentum and short term carry in interest rates. Short term mean reversion, long term volatility and long term equity momentum seem to ineffective indicators for our product set.

We try two methods with different degrees of freedom. In the first method, same indicators were shared by all currency pairs and the best combination shows an improvement in sharpe ratio by 0.1 or 20% over the best performing strategy. The best combination method based on our walk-forward simulations is maximization of the 10th percentile of rolling one year sharpe ratio while setting future weights and the method that tries to maximize the diversification of the portfolio. We demonstrate that these indicator combination strategies can work with ineffective indicators with low or negative returns over the duration of the backtest.

In the second method, we increase the search space of strategies by allowing different currency pairs to look at different indicators. Here, we find that the sharpe ratio improves further by upto 0.3 or 60% over the best performing strategy. The performance improvement in the second method confirms that different currency pairs are affected by different risk factors.

6 Further Work

While the research presented here is fairly exhaustive we believe that there a number of ways using with we can enhance portfolio construction and performance in out of sample data.

We can study the fundamental reasons for the dependence of certain currency pairs on a particular factor, be it macro-economic or based on trading patterns. Once we have identified these dynamics we can explore adding other currency pairs, specially cross pairs, as the dynamics of a particular indicator might be subdued when paired with the dollar but might be more identifiable and exploitable in a cross-pair. Converting a illiquid cross pair to liquid futures is something that should be possible with insignificant slippage.

There could many other factors or indicators that can be explored for building a foreign exchange portfolio. The ones we missed were the macro economic factors like real exchange rates, inflation, purchasing power parity etc. We can also look at particular commodities like crude, copper, gold as they might be significant in emerging markets.

We did not explicitly contain exposure to any single currency that may arise as a result of different strategies. US Dollar being the biggest currency and the fact all the traded securities are paired against USD, it is likely that an unconstrained portfolio has high exposure to US dollar which is undesirable. That is an obvious direction of future work. Moreover, any practical portfolio construction would typically employ some sort of risk management or drawdown protection.

Exploring other position sizing or changing changing existing risk budgeting based position sizing to account for different risk factors is another promising direction of future work.

While our simulation is walk-forward which leads to time varying weights, we can explore looking at alternate methods that help us decide the optimal allocations over

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time. One such method could be regime detection, where we find the best indicator or product sets with respect to regimes defined by economic variables or index levels. Srivastava et al, in [SB18] show that a using a combination strategy of Markov auto regressive models and moving averages can enhance the returns of simple strategies by adding the regime adaptive elements to them. In the paper on optimal tactical allocation to investment strategies [qpl17], the authors use a recommender systems approach for regime identification and then use it for changing allocation to strategies

We can go even further and look at non-linear methods like neural networks for indicator, strategy selection to extract more out of this strategy as the dynamics of foreign exchange markets are complex and not necessarily linear. Chakravorty et al, in [CA18] show how one can use deep neural networks with macro-economic data in conjunction with price-volume data in a walk-forward setting to do tactical asset allocation. In a follow up to this research we can add these non-linear methods and evaluate their efficacy.

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Appendix

Construction of Covariance Matrix

We break down covariance matrix into correlation and standard deviation of different securities as follows

$$\sum_{ij} = \rho_{ij} * \sigma_i * \sigma_j$$

where, $\sum_{ij}$ is covariance between i^{th} and j^{th} security,
ρ_{ij} is correlation between i^{th} and j^{th} security,
σ_i is standard deviation of the ith security.

We compute correlation values using last 1 year of realized log returns and risk or standard deviation is computed as an unweighted average of standard deviation of realized returns computed over last one month, three months and one year.

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Disclosures

The factual information set forth herein has been obtained or derived from sources believed to be reliable but it is not necessarily all-inclusive and is not guaranteed as to its accuracy and is not to be regarded as a representation or warranty, express or implied, as to the information, accuracy or completeness, nor should the attached information serve as the basis of any investment decision. Past performance is not indicative of future performance.

This document contains hypothetical performance results. Hypothetical performance results have many inherent limitations, some of which are described below. No representation is being made that any account will or is likely to achieve profits or losses similar to those shown. In fact, there are frequently sharp differences between hypothetical performance results and the actual results subsequently achieved by any particular trading program.

One of the limitations of hypothetical performance results is that they are generally prepared with the benefit of hindsight. In addition, hypothetical trading does not involve financial risk, and no hypothetical trading record can completely account for the impact of financial risk in actual trading. For example, the ability to withstand losses or adhere to a particular trading program in spite of trading losses are material points which can also adversely affect actual trading results. There are numerous other factors related to the markets in general or to the implementation of any specific trading program which cannot be fully accounted for in the preparation of hypothetical performance results and all of which can adversely affect actual trading results.

Investing in futures, derivatives or foreign exchange markets is highly speculative and involves substantial investment, liquidity and other risks. CTA managed accounts and hedge funds can be leveraged and their performance results can be volatile. Past performance of issuers, financial instruments and markets may not be indicative of future results, and there is no guarantee that targeted performance will be achieved.

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FX_MultiStrategy_V2.ipynb

Code Python 3

FX Multi Strategy - Per product indicator selection

Here for the FX futures we show -

1. Regression Strength of Indicators chosen
2. Indicator Returns
3. Combination strategies - with same indicator weights across products
4. Combination strategies - choosing indicators per product and aggregating them
5. Conclusion

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Qplum Project Management Link

[1]:
```
import datetime
import numpy as np
import pandas as pd
from qdata import DailyData
from scipy.optimize import minimize
from typing import List
import warnings
import os
warnings.filterwarnings("ignore")

#qplum imports
from stratdev.fasttest.position_sizing.risk_budgeting_position_sizing import RiskBudgetingPositionSizing
from stratdev_utils.core.db.common import get_log_returns, get_formatted_summary_stats_from_df
from stratdev_utils.performance.performance_utils import get_sharpe_ratio, get_annualized_stdev
from stratdev_utils.strategy_aggregation_modules.multiple_indicator_combination import MultipleIndicatorCombination
from stratdev_utils.strategy_aggregation_modules.strategy_aggregators import TimeSeriesAggregators
from stratdev_utils.core.output.graphical import line_chart, plot_df, bar_chart

#define static variables
start_date = datetime.date(1995, 1, 1)
end_date = datetime.date(2019,1, 20)

#this normalization technique gives a percentile score to time series values in a walkforward fashion
normalization = 'range_percentile_normalize_df_walk_forward'

#if CREATE_NEW flag is off we use the stored data - other wise we recompute everything
CREATE_NEW = 0

DATA_DIR = "/home/sonam/data/"

if not os.path.isdir(os.path.join(DATA_DIR,'fxmulti_results')): os.makedirs(os.path.join(DATA_DIR,'fxmulti_results'))
```

[4]:
```
future_to_yield1 = {
    "6A_1": "AUS10Y",
    "6B_1": "GBR10Y",
    "6C_1": "CAN10Y",
    "6E_1": "DEU10Y",
    "6J_1": "JPN10Y",
    "6M_1": "MEX10Y",
    "6N_1": "NZL10Y",
    "6S_1": "CHE10Y"
}
future_to_yield2 = {
    "6A_1": "USA10Y",
    "6B_1": "USA10Y",
    "6C_1": "USA10Y",
    "6E_1": "USA10Y",
    "6J_1": "USA10Y",
    "6M_1": "USA10Y",
    "6N_1": "USA10Y",
    "6S_1": "USA10Y"
}
future_to_spot = {
    "6A_1": "AUDUSD",
    "6B_1": "GBPUSD",
    "6C_1": "CADUSD",
    "6E_1": "EURUSD",
    "6J_1": "JPYUSD",
    "6M_1": "MXNUSD",
    "6N_1": "NZDUSD",
    "6S_1": "CHFUSD"
}
future_to_equity = {
    "6A_1": "^AXJO",
    "6B_1": "LFZ_1",
    "6C_1": "SXF_1",
    "6E_1": "EZU",
    "6J_1": "EWJ",
    "6M_1": "^MXX",
    "6N_1": "^NZ50",
    "6S_1": "^SSMI"
}

products_info = pd.concat(
    [
        pd.Series(future_to_yield1),
        pd.Series(future_to_yield2),
        pd.Series(future_to_spot),
        pd.Series(future_to_equity)
    ],
    axis=1)
products_info.columns = ['yield1', 'yield2', 'spot', 'equity']

#short term horizon is set as 3 months
    axis=1)
products_info.columns = ['yield1', 'yield2', 'spot', 'equity']

#short term horizon is set as 3 months
SHORT_TERM_HORIZON = 63
#long term horizon is set as 1 year
LONG_TERM_HORIZON = 252

display(products_info)
```

	yield1	yield2	spot	equity
6A_1	AUS10Y	USA10Y	AUDUSD	^AXJO
6B_1	GBR10Y	USA10Y	GBPUSD	LFZ_1
6C_1	CAN10Y	USA10Y	CADUSD	SXF_1
6E_1	DEU10Y	USA10Y	EURUSD	EZU
6J_1	JPN10Y	USA10Y	JPYUSD	EWJ
6M_1	MEX10Y	USA10Y	MXNUSD	^MXX
6N_1	NZL10Y	USA10Y	NZDUSD	^NZ50
6S_1	CHE10Y	USA10Y	CHFUSD	^SSMI

The strategy class definition

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[5]:
```
class FX_MultiStrategy(MultipleIndicatorCombination):
    """
    This class inherits the indicator combination, scaling etc functionality from MultipleIndicatorCombination
    and implements the combination strategy for FX futures - using momenum, mean reversion, yield diff,
    equity momentum, commodity momentum
    """

    def __init__(self,
                 start_date: datetime.date,
                 end_date: datetime.date,
                 products_info: pd.DataFrame,
                 normalization: str) -> None:
        """
        initialize the class
        :param :
        """
        # start date is minimum of start date and a date 6 years from now - as the indicator uses
        # 5 years of data to train
        start_date = min(datetime.date.today() - datetime.timedelta(days=252 * 6), start_date)
        rba_config = {
            'risk_config': {
                'correlation': {
                    'lookback': 252
                },
                'correlation': {
                    'lookback': 252
                },
                'risk_model': {
                    'average_type': 'exponential',
                    'name': 'AverageStdevRiskModel',
                    'periods': [21, 63, 252]
                }
            },
            'previous_weights': None,
            'maximum_leverage': 4.0,
            'minimum_allocation': -3.0,
            'maximum_allocation': 3.0,
            'target_risk': 10.0,
            'display_opt_output': False,
            'initialize_with_previous_weights': 0,
            'marketcap_method': {
                'sector_weight_map': {'Currency': 1},
                'name': 'constant'
            },
            'start_date': start_date,
            'end_date': end_date,
            'product_list': list(products_info.index)
        }
        super().__init__(start_date, end_date, normalization, RiskBudgetingPositionSizing, rba_config)
        self._products_info = products_info
        prod_list = list(self._products_info.index)
        prod_list.extend([c.replace('_1', '_2') for c in prod_list])
        self._logret_df = get_log_returns(prod_list, self._start_date, self._end_date)
        self._cum_logret_df = self._logret_df.cumsum()
        self._equity_logret_df = get_log_returns(
            list(self._products_info.equity), self._start_date, self._end_date)
        self._commodity_logret_df = get_log_returns(
            ['^SPGSCI', 'BRN_1', 'CL_1', 'GC_1',
             'HG_1', 'DBA'], self._start_date, self._end_date)
        self._create_all_indicators()
        self._normalize_and_scale_all_indicators(parallelize=0)
        self._convert_all_indicators_to_returns(parallelize=0)

    def _raw_momentum(self, currency_future_ticker: str,
                      n: int) -> pd.Series:
        """
        returns n day momentum for the giver fx future ticker
        """
        return self._logret_df[currency_future_ticker].rolling(n).mean()/self._logret_df[currency_future_ticker].rolling(n).std()

    def _raw_mean_reversion(self, currency_future_ticker: str,
                            n: int) -> pd.Series:
        """
        returns n day mean reversion for the giver fx future ticker
        """
        return 1 + (self._cum_logret_df[currency_future_ticker].rolling(n).mean() - self._cum_logret_df[currency_future_ticker])/self._cum_logret_df[currency_future_ticker].rolling(n).std()
        US_yield = US_yield[US_yield.index >= self._start_date]
        return ((1 + local_yield / 100.0) /
                (1 + US_yield / 100.0)).apply(lambda x: np.log(x))
    def _raw_short_term_yield_diff(self, currency_future_ticker: str) -> pd.Series:
        """
        returns yield difference for the given fx future ticker
        """
        try:
            yield1 = self._products_info.loc[currency_future_ticker, 'yield1'].replace('10Y', '3M')
            yield2 = self._products_info.loc[currency_future_ticker, 'yield2'].replace('10Y', '3M')
            local_yield = DailyData([yield1]).raw_data().stack(
                level=0).reset_index().set_index('date')['rate'].apply(
                    lambda x: float(x))
            local_yield = local_yield[local_yield.index >= self._start_date]
            US_yield = DailyData([yield2]).raw_data().stack(
                level=0).reset_index().set_index('date')['rate'].apply(
                    lambda x: float(x))
            US_yield = US_yield[US_yield.index >= self._start_date]
            return ((1 + local_yield / 100.0) /
                    (1 + US_yield / 100.0)).apply(lambda x: np.log(x))
        except:
            yield1 = self._products_info.loc[currency_future_ticker, 'yield1'].replace('10Y', '1Y')
            yield2 = self._products_info.loc[currency_future_ticker, 'yield2'].replace('10Y', '1Y')
            local_yield = DailyData([yield1]).raw_data().stack(
                level=0).reset_index().set_index('date')['rate'].apply(
                    lambda x: float(x))
            local_yield = local_yield[local_yield.index >= self._start_date]
            US_yield = DailyData([yield2]).raw_data().stack(
                level=0).reset_index().set_index('date')['rate'].apply(
                    lambda x: float(x))
            US_yield = US_yield[US_yield.index >= self._start_date]
            return ((1 + local_yield / 100.0) /
                    (1 + US_yield / 100.0)).apply(lambda x: np.log(x))
    def _raw_equity_momentum(self, currency_future_ticker: str,
                             n: int) -> pd.Series:
        """
        returns n day mometum for the equity index linked to given fx future ticker
        """
        equity_ticker = self._products_info.loc[currency_future_ticker,
                                                'equity']
        return self._equity_logret_df[equity_ticker].rolling(n).mean()/self._equity_logret_df[equity_ticker].rolling(n).std()
    # noinspection PyUnusedLocal
    def _raw_commodity_momentum(self, currency_future_ticker: str,
                                n: int) -> pd.Series:
        """
        returns n day mometum for the commodity index linked to given fx future ticker
        """
        commodity_ticker = '^SPGSCI'
        return self._commodity_logret_df[commodity_ticker].rolling(n).mean()/self._commodity_logret_df[commodity_ticker].rolling(n).std()
    def _raw_gold_momentum(self, currency_future_ticker: str,
                                n: int) -> pd.Series:
        """
        returns n day mometum for the commodity index linked to given fx future ticker
        """
        commodity_ticker = 'BRN_1'
        return self._commodity_logret_df[commodity_ticker].rolling(n).mean()/self._commodity_logret_df[commodity_ticker].rolling(n).std()
    def _raw_agri_momentum(self, currency_future_ticker: str,
                                n: int) -> pd.Series:
        """
        returns n day mometum for the commodity index linked to given fx future ticker
        """
        commodity_ticker = 'DBA'
        return self._commodity_logret_df[commodity_ticker].rolling(n).mean()/self._commodity_logret_df[commodity_ticker].rolling(n).std()

    def _raw_carry(self, currency_future_ticker: str):
```

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```
    def _raw_carry(self, currency_future_ticker: str,
                   n: int) -> pd.Series:
        return self._logret_df[currency_future_ticker].rolling(n).sum() \
               - self._logret_df[currency_future_ticker.replace('_1','_2')].rolling(n).sum()

    def _raw_indicator(self, type_: str, n: int) -> pd.Series:
        """
        Here we get raw data of mean reversion, momentum or yield into the indicator df
        """
        indicators = pd.DataFrame()
        for k in self._products_info.index:
            x = None
            if type_ == 'momentum':
                x = self._raw_momentum(k, n)
            elif type_ == 'mean_reversion':
                x = self._raw_mean_reversion(k, n)
            elif type_ == 'short_term_yield_diff':
                x = self._raw_short_term_yield_diff(k)
            elif type_ == 'long_term_yield_diff':
                x = self._raw_long_term_yield_diff(k)
            elif type_ == 'vol':
                x = self._raw_volatility(k, n)
            elif type_ == 'equity_mom':
                x = self._raw_equity_momentum(k, n)
            elif type_ == 'commodity_mom':
                x = self._raw_commodity_momentum(k, n)
            elif type_ == 'gold_mom':
                x = self._raw_gold_momentum(k, n)
            elif type_ == 'crude_mom':
                x = self._raw_crude_momentum(k, n)
            elif type_ == 'agri_mom':
                x = self._raw_agri_momentum(k, n)
            elif type_ == 'carry':
                x = self._raw_carry(k, n)
            if indicators.empty:
                indicators = x.to_frame(k)
            else:
                indicators = pd.concat(
                    [indicators, x.to_frame(k)], axis=1, join='outer')
        indicators.index = pd.to_datetime(indicators.index).date
        x = None
            if type_ == 'momentum':
                x = self._raw_momentum(k, n)
            elif type_ == 'mean_reversion':
                x = self._raw_mean_reversion(k, n)
            elif type_ == 'short_term_yield_diff':
                x = self._raw_short_term_yield_diff(k)
            elif type_ == 'long_term_yield_diff':
                x = self._raw_long_term_yield_diff(k)
            elif type_ == 'vol':
                x = self._raw_volatility(k, n)
            elif type_ == 'equity_mom':
                x = self._raw_equity_momentum(k, n)
            elif type_ == 'commodity_mom':
                x = self._raw_commodity_momentum(k, n)
            elif type_ == 'gold_mom':
                x = self._raw_gold_momentum(k, n)
            elif type_ == 'crude_mom':
                x = self._raw_crude_momentum(k, n)
            elif type_ == 'agri_mom':
                x = self._raw_agri_momentum(k, n)
            elif type_ == 'carry':
                x = self._raw_carry(k, n)
            if indicators.empty:
                indicators = x.to_frame(k)
            else:
                indicators = pd.concat(
                    [indicators, x.to_frame(k)], axis=1, join='outer')
        indicators.index = pd.to_datetime(indicators.index).date
        return indicators

    def _create_all_indicators(self) -> None:
        """
        create all indicator dataframes for all products
        :return:
        """
        self._all_indicators['short_term_yield_diff'] = self._raw_indicator(
            type_='short_term_yield_diff', n=0)
        self._all_indicators['long_term_yield_diff'] = self._raw_indicator(
            type_='long_term_yield_diff', n=0)
        self._all_indicators['short_term_momentum'] = self._raw_indicator(
            type_='momentum', n=SHORT_TERM_HORIZON)
        self._all_indicators['long_term_momentum'] = self._raw_indicator(
            type_='momentum', n=LONG_TERM_HORIZON)
        self._all_indicators[
            'short_term_mean_reversion'] = self._raw_indicator(
                type_='mean_reversion', n=SHORT_TERM_HORIZON)
        self._all_indicators['long_term_mean_reversion'] = self._raw_indicator(
            type_='mean_reversion', n=LONG_TERM_HORIZON)
        self._all_indicators['short_term_volatility'] = self._raw_indicator(
            type_='vol', n=SHORT_TERM_HORIZON)
        self._all_indicators['long_term_volatility'] = self._raw_indicator(
        self._all_indicators[
            'short_term_equity_momentum'] = self._raw_indicator(
                type_='equity_mom', n=SHORT_TERM_HORIZON)
        self._all_indicators[
            'long_term_equity_momentum'] = self._raw_indicator(
                type_='equity_mom', n=LONG_TERM_HORIZON)
        self._all_indicators[
            'short_term_commodity_momentum'] = self._raw_indicator(
                type_='commodity_mom', n=SHORT_TERM_HORIZON)
        self._all_indicators[
            'long_term_commodity_momentum'] = self._raw_indicator(
                type_='commodity_mom', n=LONG_TERM_HORIZON)
        '''
        self._all_indicators[
            'short_term_carry'] = self._raw_indicator(
                type_='carry', n=SHORT_TERM_HORIZON)
        self._all_indicators[
            'long_term_carry'] = self._raw_indicator(
                type_='carry', n=LONG_TERM_HORIZON)
        self._all_indicators[
            'short_term_crude_momentum'] = self._raw_indicator(
                type_='crude_mom', n=SHORT_TERM_HORIZON)
        self._all_indicators[
            'long_term_crude_momentum'] = self._raw_indicator(
                type_='crude_mom', n=LONG_TERM_HORIZON)
        self._all_indicators[
            'short_term_gold_momentum'] = self._raw_indicator(
                type_='gold_mom', n=SHORT_TERM_HORIZON)
        self._all_indicators[
            'long_term_gold_momentum'] = self._raw_indicator(
                type_='gold_mom', n=LONG_TERM_HORIZON)
        self._all_indicators[
            'short_term_agri_momentum'] = self._raw_indicator(
                type_='agri_mom', n=SHORT_TERM_HORIZON)
        self._all_indicators[
            'long_term_agri_momentum'] = self._raw_indicator(
                type_='agri_mom', n=LONG_TERM_HORIZON)
        '''
```

[6]:
```
fxmulti = FX_MultiStrategy(start_date, end_date, products_info, normalization)
fxmulti.fname_dict.pop('Max_Conservative_Sharpe')
```

[6]:
```
<bound method TimeSeriesAggregators._max_conservative_sharpe_optimal_portfolio of <__main__.FX_MultiStrategy object at 0x7fea30e45ba8>>
```

Regression Results

Here we look at tstats and p values to evaluate the strength of regression

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Here we look at tstats and p-values to evaluate the strength of regression

```
                    ])

        self.t = self.coef_ / se
        self.p = 2 * (1 - scipy_stats.t.cdf(np.abs(self.t), y.shape[0] - X.shape[1]))
        return self
```

[9]:
```
tstats = pd.DataFrame(index = fxmulti._products_info.index, columns = fxmulti._all_indicators.keys())
pstats = pd.DataFrame(index = fxmulti._products_info.index, columns = fxmulti._all_indicators.keys())
for k in fxmulti._products_info.index:
    for i in fxmulti._all_indicators.keys():
        Y = fxmulti._future_return[k]
        X = fxmulti._all_indicators[i][k].fillna(method='ffill')
        data = pd.concat([X, Y], axis=1)
        data.columns = ['X', 'Y']
        data.dropna(inplace=True)
        X = data['X'].to_frame()
        Y = data['Y'].to_frame()
        reg = LinearRegression().fit(X, Y)
        stars = '***'
        if reg.p[0][0] < 0.01:
            statrs ='***'
        elif reg.p[0][0] < 0.05:
            stars = '**'
        elif reg.p[0][0] < 0.1:
            stars = '*'
        else:
            stars = ''
        tstats.loc[k, i] = '{}{}'.format(np.round(reg.t[0][0],3), stars)
        pstats.loc[k, i] = '({})'.format(np.round(reg.p[0][0], 3))
```

Tstats of the regression to predict future values using single indicators

[10]:
```
tstats.columns = tstats.columns.str.replace('_', ' ').str.title()
display(tstats)
```

	Short Term Yield Diff	Long Term Yield Diff	Short Term Momentum	Long Term Momentum	Short Term Mean Reversion	Long Term Mean Reversion	Short Term Volatility	Long Term Volatility	Short Term Equity Momentum	Long Term Equity Momentum	Short Term Commodity Momentum	Long Term Commodity Momentum
6A_1	15.581***	15.522***	52.867***	31.488***	-61.359***	-39.406***	6.814***	9.899***	13.501***	2.689***	23.096***	11.694***
6B_1	7.022***	8.084***	50.686***	21.169***	-62.374***	-35.299***	1.206	3.052***	5.599***	4.513***	16.042***	2.538**
6C_1	8.81***	10.102***	55.09***	28.401***	-66.192***	-38.031***	1.899*	2.547**	12.412***	5.905***	25.565***	12.839***
6E_1	-1.14	10.132***	53.27***	24.593***	-62.655***	-36.335***	0.346	1.41	19.353***	7.027***	15.706***	6.152***
6J_1	2.641***	13.905***	61.359***	33.413***	-78.392***	-45.106***	-8.372***	-9.377***	3.892***	3.28***	1.19	-1.991**
6M_1	10.28***	6.07***	58.013***	34.334***	-55.057***	-32.762***	16.027***	20.952***	25.237***	21.042***	15.406***	19.095***
6N_1	9.355***	12.731***	52.551***	29.697***	-58.356***	-35.063***	7.032***	9.967***	12.805***	9.637***	17.812***	4.395***
6S_1	9.988***	9.237***	51.641***	28.678***	-67.856***	-39.856***	-5.529***	-5.578***	-10.626***	-11.345***	10.684***	2.843***


```
            stars = '**'
        elif reg.p[0][0] < 0.1:
            stars = '*'
        else:
            stars = ''
        tstats.loc[k, i] = '{}{}'.format(np.round(reg.t[0][0],3), stars)
        pstats.loc[k, i] = '({})'.format(np.round(reg.p[0][0], 3))
```

Tstats of the regression to predict future values using single indicators

[10]:
```
tstats.columns = tstats.columns.str.replace('_', ' ').str.title()
display(tstats)
```

	Short Term Yield Diff	Long Term Yield Diff	Short Term Momentum	Long Term Momentum	Short Term Mean Reversion	Long Term Mean Reversion	Short Term Volatility	Long Term Volatility	Short Term Equity Momentum	Long Term Equity Momentum	Short Term Commodity Momentum	Long Term Commodity Momentum
6A_1	15.581***	15.522***	52.867***	31.488***	-61.359***	-39.406***	6.814***	9.899***	13.501***	2.689***	23.096***	11.694***
6B_1	7.022***	8.084***	50.686***	21.169***	-62.374***	-35.299***	1.206	3.052***	5.599***	4.513***	16.042***	2.538**
6C_1	8.81***	10.102***	55.09***	28.401***	-66.192***	-38.031***	1.899*	2.547**	12.412***	5.905***	25.565***	12.839***
6E_1	-1.14	10.132***	53.27***	24.593***	-62.655***	-36.335***	0.346	1.41	19.353***	7.027***	15.706***	6.152***
6J_1	2.641***	13.905***	61.359***	33.413***	-78.392***	-45.106***	-8.372***	-9.377***	3.892***	3.28***	1.19	-1.991**
6M_1	10.28***	6.07***	58.013***	34.334***	-55.057***	-32.762***	16.027***	20.952***	25.237***	21.042***	15.406***	19.095***
6N_1	9.355***	12.731***	52.551***	29.697***	-58.356***	-35.063***	7.032***	9.967***	12.805***	9.637***	17.812***	4.395***
6S_1	9.988***	9.237***	51.641***	28.678***	-67.856***	-39.856***	-5.529***	-5.578***	-10.626***	-11.345***	10.684***	2.843***

Pvalues of the regression to predict future values using single indicators

[11]:
```
pstats.columns = pstats.columns.str.replace('_', ' ').str.title()
display(pstats)
```

	Short Term Yield Diff	Long Term Yield Diff	Short Term Momentum	Long Term Momentum	Short Term Mean Reversion	Long Term Mean Reversion	Short Term Volatility	Long Term Volatility	Short Term Equity Momentum	Long Term Equity Momentum	Short Term Commodity Momentum	Long Term Commodity Momentum
6A_1	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.007)	(0.0)	(0.0)
6B_1	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.228)	(0.002)	(0.0)	(0.0)	(0.0)	(0.011)
6C_1	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.058)	(0.011)	(0.0)	(0.0)	(0.0)	(0.0)
6E_1	(0.254)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.73)	(0.159)	(0.0)	(0.0)	(0.0)	(0.0)
6J_1	(0.008)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.001)	(0.234)	(0.046)
6M_1	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
6N_1	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)
6S_1	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.0)	(0.004)

Indicator Returns

Returns of individual indicators (across products)

[12]:
```
if CREATE_NEW:
    def fn(indicators, name):
        logret_df = fxmulti._combine_indicators_using_rba(indicators)
        logret_df /= get_annualized_stdev(logret_df)/10
        stats = get_formatted_summary_stats_from_df(logret_df)
        return logret_df['logret'].to_frame(name), stats['logret'].to_frame(name)

    logret = pd.DataFrame()
    stats = pd.DataFrame()
    for indicator in fxmulti._all_indicators_scaled:
        indicators = fxmulti._all_indicators_scaled[indicator]
        lr, st = fn(indicators, indicator)
        logret = pd.concat([logret, lr], axis=1)
        stats = pd.concat([stats, st], axis=1)
```

[13]:
```
if CREATE_NEW:
    logret.to_csv(os.path.join(DATA_DIR,'fxmulti_results/indicator_returns.csv'))
    stats.to_csv(os.path.join(DATA_DIR,'fxmulti_results/indicator_stats.csv'))
else:
    logret = pd.read_csv(os.path.join(DATA_DIR,'fxmulti_results/indicator_returns.csv'), index_col=0, parse_dates=True)
    logret.index = logret.index.date
    stats = pd.read_csv(os.path.join(DATA_DIR,'fxmulti_results/indicator_stats.csv'), index_col=0)
```

Stats for strategies created using single indicators

[14]:
```
stats.columns = [s.replace('_', ' ').title() for s in stats.columns]
stats.loc[['annualized_returns', 'annual_sharpe_ratio_daily_returns'], :].T.sort_values('annual_sharpe_ratio_daily_returns', ascending=False)
```

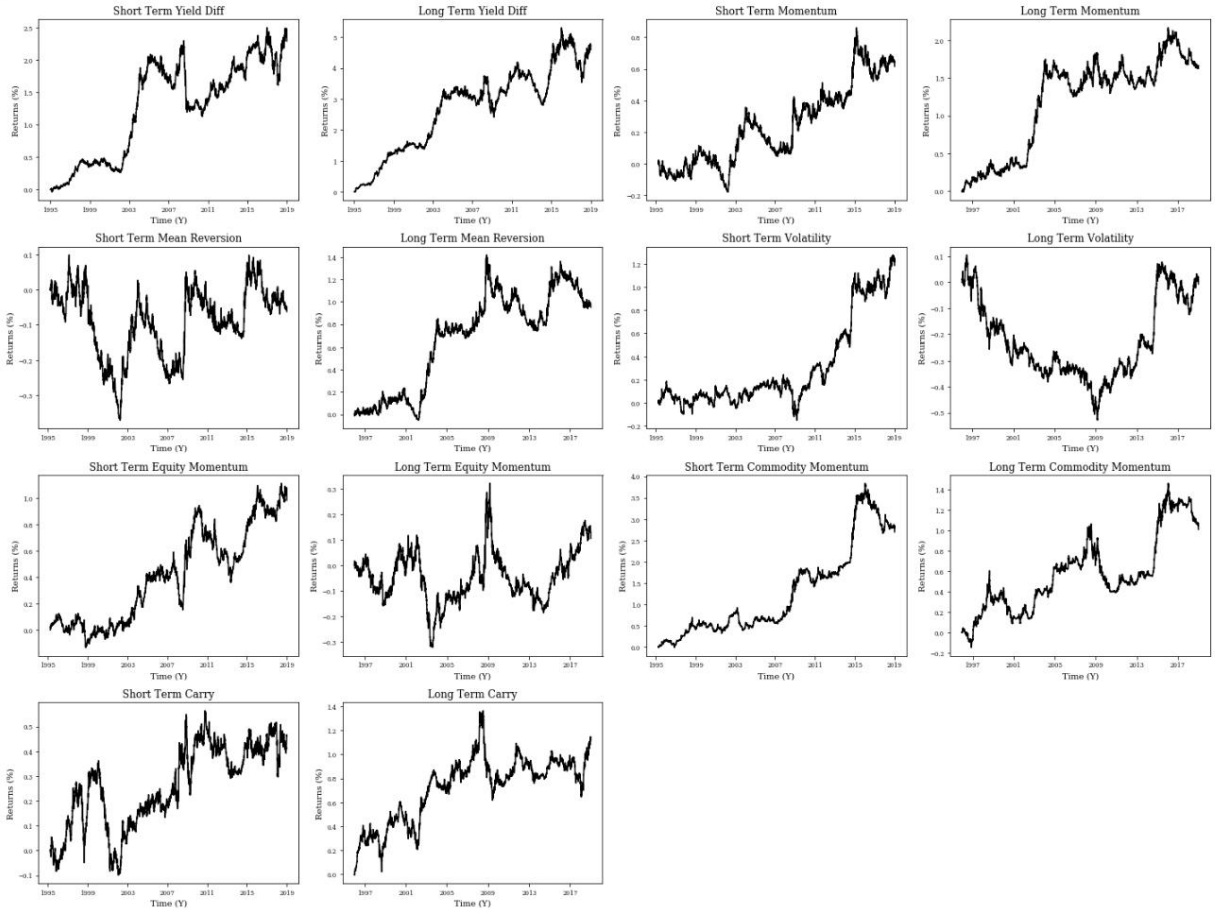
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[14]:

stat	annualized_returns	annual_sharpe_ratio_daily_returns
Long Term Yield Diff	7.45	0.54
Short Term Commodity Momentum	5.67	0.37
Short Term Yield Diff	5.11	0.31
Long Term Momentum	4.30	0.24
Short Term Volatility	3.37	0.15
Long Term Carry	3.31	0.15
Long Term Commodity Momentum	3.09	0.12
Long Term Mean Reversion	2.95	0.11
Short Term Equity Momentum	2.94	0.10
Short Term Momentum	2.05	0.02
Short Term Carry	1.62	-0.03
Long Term Equity Momentum	0.44	-0.14
Long Term Volatility	-0.03	-0.18
Short Term Mean Reversion	-0.27	-0.21

Charts for strategies created using single indicators

[16]:
```
import matplotlib.pyplot as plt
from stratdev.performance.performance_utils import *
plt.rc('font', family='serif')
plt.rc('xtick', labelsize='x-small')
plt.rc('ytick', labelsize='x-small')
lr = logret.copy()
fig = plt.figure(figsize=(20, 15))
for i in range(14):
    ax = fig.add_subplot(4, 4, i+1)
    x = lr[lr.columns[i]].cumsum().apply(log_to_nominal)
    ax.plot(x,  color='k', ls='solid')
    ax.set_xlabel('Time (Y)')
    ax.set_ylabel('Returns (%)')
    plt.title(lr.columns[i].replace('_', ' ').title())
    plt.tight_layout()
plt.show()
```



[18]:
```
all_stats = {}
for ind in fxmulti._all_returns:
    all_stats[ind] = get_formatted_summary_stats_from_df(fxmulti._all_returns[ind].fillna(0))
```

[19]:
```
ann_returns = pd.DataFrame()
for k in all_stats:
    ann_returns=pd.concat([ann_returns,all_stats[k].loc['annualized_returns', :].T], axis=1)
```

[20]:
```
ann_returns.columns = all_stats.keys()
```

Correlation between indicator based strategies

[22]:
```
logret.corr().round(2).style.background_gradient(cmap='summer_r')
```

[22]:

	short_term_yield_diff	long_term_yield_diff	short_term_momentum	long_term_momentum	short_term_mean_reversion	long_term_mean_reversion	short_term_volatility	long_term_volatility	short_term_equity_momentum	long_term_equ
short_term_yield_diff	1	0.65	0.2	0.31	0.14	0.28	0.37	0.36	-0.15	
long_term_yield_diff	0.65	1	0.27	0.42	0.23	0.39	0.24	0.25	-0.08	
short_term_momentum	0.2	0.27	1	0.56	0.83	0.76	-0.02	0.01	0.26	
long_term_momentum	0.31	0.42	0.56	1	0.5	0.83	-0.08	-0.07	0.13	
short_term_mean_reversion	0.14	0.23	0.83	0.5	1	0.71	-0.02	0.01	0.26	
long_term_mean_reversion	0.28	0.39	0.76	0.83	0.71	1	-0.05	-0.02	0.22	
short_term_volatility	0.37	0.24	-0.02	-0.08	-0.02	-0.05	1	0.7	-0.1	
long_term_volatility	0.36	0.25	0.01	-0.07	0.01	-0.02	0.7	1	-0.07	
short_term_equity_momentum	-0.15	-0.08	0.26	0.13	0.26	0.22	-0.1	-0.07	1	
long_term_equity_momentum	-0.21	-0.19	0.09	0.2	0.09	0.18	-0.27	-0.21	0.52	
short_term_commodity_momentum	0.06	0.12	0.38	0.32	0.39	0.38	0.02	0.03	0.28	

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long_term_commodity_momentum	0.16	0.21	0.18	0.43	0.17	0.32	-0.06	-0.02	0.07
short_term_carry	0.16	0.09	0.09	-0.1	0.1	0.01	0.14	0.11	-0.05
long_term_carry	0.36	0.3	0.09	0.04	0.08	0.12	0.19	0.2	-0.18

Clustering currencies based on indicator retruns

[23]:
```
from sklearn.cluster import KMeans
kmeans = KMeans(n_clusters=2, random_state=0).fit(ann_returns)
kmeans.labels_
```

[23]:
```
array([0, 1, 0, 0, 0, 0, 0, 1], dtype=int32)
```

[24]:
```
if CREATE_NEW: fxmulti.compute_all_indicators_insample(parallelize=1)
```

[25]:
```
if CREATE_NEW:fxmulti.compute_all_indicators_outsample(parallelize=1)
```

[26]:
```
if CREATE_NEW:
    for method in  fxmulti._final_indicator_df_insample.keys():
        tsmom._final_indicator_df_insample[method].to_csv(os.path.join(DATA_DIR,'fxmulti_results/insample/'+method+'_all.csv'))
        tsmom._final_indicator_df_walkforward[method].to_csv(os.path.join(DATA_DIR,'fxmulti_results/outsample/'+method+'_all.csv'))
```

1. Returns of Combination Stratgies

Combining such that same weight is added to the indicators across products

[27]:
```
if CREATE_NEW:
    import os
    from stratdev.performance.performance_utils import get_annualized_stdev
    log_returns_df = pd.DataFrame()
    stats_df = pd.DataFrame()
    for directory in [os.path.join(DATA_DIR,'fxmulti_results/insample'), os.path.join(DATA_DIR,'fxmulti_results/outsample')]:
        for f in os.listdir(directory):
            indicators = pd.read_csv(directory+'/'+f, index_col=0, parse_dates=True)
            name = f.replace('.csv', '')+directory.replace('results/','_')
            print (name)
            logret_df = fxmulti._combine_indicators_using_rba(indicators)
            logret_df /= get_annualized_stdev(logret_df)/10
            stats = get_formatted_summary_stats_from_df(logret_df)
            log_returns_df = pd.concat([log_returns_df, logret_df['logret'].to_frame(name)], axis=1)
            stats_df = pd.concat([stats_df, stats['logret'].to_frame(name)], axis=1)
```

[28]:
```
if CREATE_NEW:
    log_returns_df.to_csv(os.path.join(DATA_DIR,'fxmulti_results/log_returns_df.csv'))
    stats_df.to_csv(os.path.join(DATA_DIR,'fxmulti_results/stats_df.csv'))
else:
    log_returns_df = pd.read_csv(os.path.join(DATA_DIR,'fxmulti_results/log_returns_df.csv'), index_col=0, parse_dates=True)
    log_returns_df.index = log_returns_df.index.date
    stats_df = pd.read_csv(os.path.join(DATA_DIR,'fxmulti_results/stats_df.csv'), index_col=0)
```

Stats for the combination strategies

[29]:
```
sdf = stats_df.loc[:, [c for c in stats_df.columns if 'outsample' in c ]]
sdf.columns = [x.replace('_all_outsample','').replace('_', ' ').title() for x in sdf.columns]
sdf = sdf.rename({'Invcorrelation By Sigma': 'Volatility scaled negative correlation',
                  'Inverse Correlation Dot Sharpe':'Negative correlation dot Sharpe',
                 'Max Rolling 10 Pctile Sharpe':'Optimization to maximize 10th percentile of Sharpe',
                 'Optimize Sharpe':'Optimization to maximize Sharpe',
                 'Max Diversification':'Max diversification',
                 'Equal Risk':'Equal risk',
                 'Equal Weight':'Equal weight',
                 'Proportional To Sharpe':'Proportional to Sharpe',
                 'Equal Risk Contribution':'Equal risk contribution'}, axis=1)
sdf.loc[['annualized_returns', 'annual_sharpe_ratio_daily_returns'], :].T.sort_values('annual_sharpe_ratio_daily_returns', ascending=False)
```

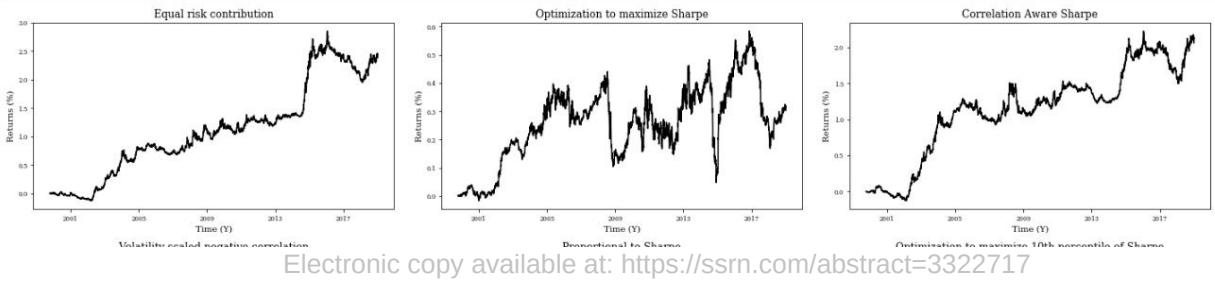
[29]:

stat	annualized_returns	annual_sharpe_ratio_daily_returns
Volatility scaled negative correlation	8.24	0.67
Max diversification	7.61	0.60
Equal risk	6.79	0.52
Equal risk contribution	6.57	0.50
Equal weight	6.58	0.50
Correlation Aware Sharpe	6.02	0.45
Proportional to Sharpe	5.91	0.44
Optimization to maximize 10th percentile of Sharpe	5.02	0.35
Negative correlation dot Sharpe	1.50	0.00
Optimization to maximize Sharpe	1.39	-0.01

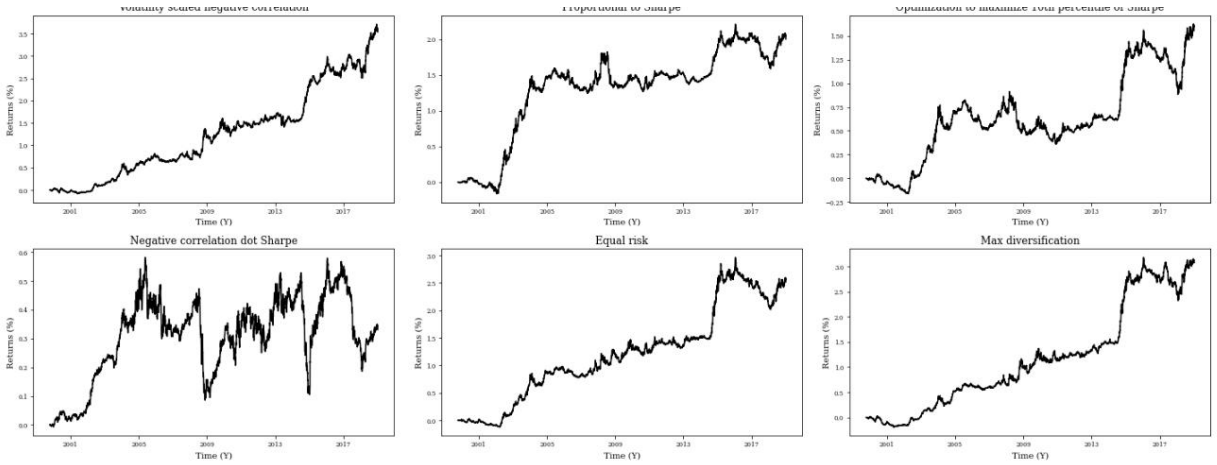
Charts for the combination strategies

[30]:
```
import matplotlib.pyplot as plt
from stratdev.performance.performance_utils import *
plt.rc('font', family='serif')
plt.rc('xtick', labelsize='x-small')
plt.rc('ytick', labelsize='x-small')
lr = log_returns_df.copy()
lr = lr.loc[:, [c for c in stats_df.columns if 'outsample' in c ]]
lr=lr.drop('Equal_Weight_all_outsample', axis=1)
lr.columns = [c.replace('_all_outsample','').replace('_', ' ').title() for c in lr.columns]
lr = lr.rename({'Invcorrelation By Sigma': 'Volatility scaled negative correlation',
                  'Inverse Correlation Dot Sharpe':'Negative correlation dot Sharpe',
                 'Max Rolling 10 Pctile Sharpe':'Optimization to maximize 10th percentile of Sharpe',
                 'Optimize Sharpe':'Optimization to maximize Sharpe',
                 'Max Diversification':'Max diversification',
                 'Equal Risk':'Equal risk',
                 'Equal Weight':'Equal weight',
                 'Proportional To Sharpe':'Proportional to Sharpe',
                 'Equal Risk Contribution':'Equal risk contribution'}, axis=1)
fig = plt.figure(figsize=(20, 15))
for i in range(9):
    ax = fig.add_subplot(4, 3, i+1)
    x = lr[lr.columns[i]].cumsum().apply(log_to_nominal)
    ax.plot(x,  color='k', ls='solid')
    ax.set_xlabel('Time (Y)')
    ax.set_ylabel('Returns (%)')
    plt.title(lr.columns[i])#.replace('_all_outsample','').replace('_', ' ').title())
    plt.tight_layout()

plt.show()
```



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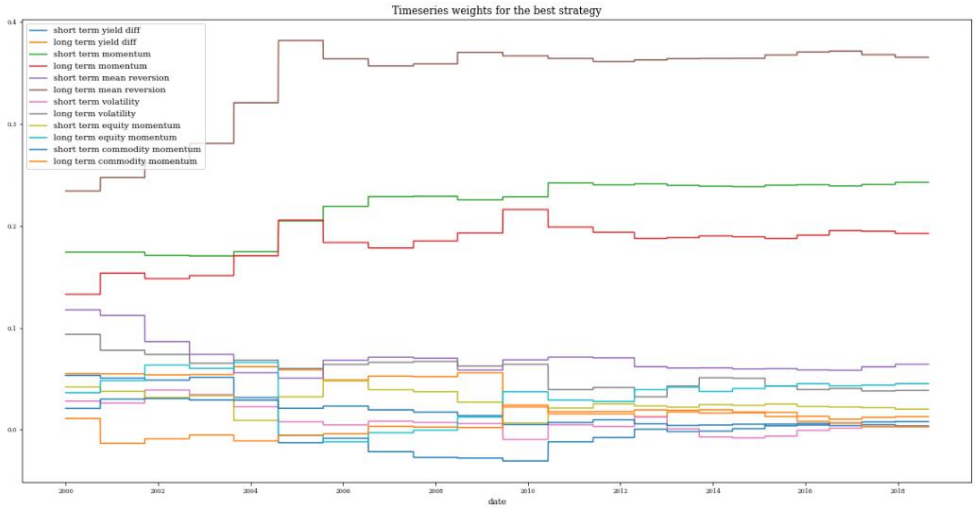


[31]:
```
a=fxmulti._get_final_indicator(method = 'InvCorrelation_By_Sigma', insample=False)
```

Weights for the best strategy - timeseries, latest and average

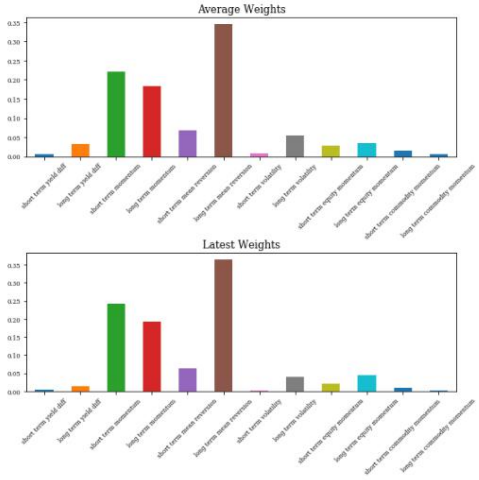
[33]:
```
a[1].columns = [c.replace('_',' ') for c in a[1].columns]
plt.figure(figsize=[20,10])
ax=plt.subplot()
a[1][a[1].index>=datetime.date(2000,1,1)].plot(ax=ax)
plt.title('Timeseries weights for the best strategy')
```

[33]:
```
Text(0.5,1,'Timeseries weights for the best strategy')
```



[34]:
```
wts = a[1][a[1].index>=datetime.date(2000,1,1)].dropna()
plt.figure(figsize=[8,8])
ax=plt.subplot(211)
wts.mean().plot(kind='bar', ax=ax, rot=45)
plt.title('Average Weights')
ax=plt.subplot(212)
wts.loc[wts.index[-1], :].plot(kind='bar', ax=ax, rot=45)
plt.tight_layout()
plt.title('Latest Weights')
```

[34]:
```
Text(0.5,1,'Latest Weights')
```



2. Combination after selecting Indicators per Product

[35]:
```
if CREATE_NEW:
    single_fx = {}
    for k in fxmulti._products_info.index:
        pinfo = products_info.loc[k, :].to_frame().T
        ifx =  FX_MultiStrategy(start_date, end_date, pinfo, normalization)
        ifx.fname_dict.pop('Max_Conservative_Sharpe')
        single_fx[k] = ifx
```

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[36]:
```
if CREATE_NEW:
    indicators = pd.DataFrame()
    returns = pd.DataFrame()
    for p in single_fx.keys():
        t = single_fx[p]
        for k in t._all_indicators_scaled.keys():
            indicators=pd.concat([indicators, t._all_indicators_scaled[k][p].to_frame(p+'_'+k)], axis=1)
            returns = pd.concat([returns, t._all_returns[k][p].to_frame(p+'_'+k)], axis=1)
    returns.to_csv(os.path.join(DATA_DIR,'fxmulti_results/indicator_product_returns.csv'))
else:
    returns = pd.read_csv(os.path.join(DATA_DIR,'fxmulti_results/indicator_product_returns.csv'),index_col=0,parse_dates=True)
    returns.index = pd.to_datetime(returns.index).date
```

[37]:
```
stats = get_formatted_summary_stats_from_df(returns.fillna(0))
avg_stats = stats
df = avg_stats.T
df = df[df.annualized_returns > 2]
print (df.index)
data =returns.loc[:, df.index]
agg = TimeSeriesAggregators(data)
agg.fname_dict.pop('Max_Conservative_Sharpe')
```

```
Index(['6A_1_long_term_mean_reversion', '6A_1_long_term_momentum',
       '6A_1_long_term_yield_diff', '6A_1_short_term_carry',
       '6B_1_long_term_equity_momentum', '6B_1_short_term_carry',
       '6B_1_short_term_equity_momentum', '6C_1_short_term_equity_momentum',
       '6C_1_short_term_volatility', '6E_1_long_term_yield_diff',
       '6J_1_long_term_volatility', '6J_1_short_term_commodity_momentum',
       '6J_1_short_term_momentum', '6J_1_short_term_volatility',
       '6M_1_short_term_yield_diff', '6N_1_long_term_yield_diff',
       '6S_1_long_term_carry', '6S_1_long_term_yield_diff',
       '6S_1_short_term_commodity_momentum', '6S_1_short_term_yield_diff'],
      dtype='object')
```

[37]:
```
<bound method TimeSeriesAggregators._max_conservative_sharpe_optimal_portfolio of <stratdev_utils.strategy_aggregation_modules.strategy_aggregators.TimeSeriesAggregators object at 0x7fe9a7f61240>>
```

[38]:
```
if CREATE_NEW:
    wto, lro = agg.compute_all_aggregators_outsample()
    lro/=get_formatted_summary_stats_from_df(lro.fillna(0)).loc['annualized_stdev', :]/10.0
    lro.to_csv(os.path.join(DATA_DIR,'fxmulti_results/logret_gran_out.csv'))
else:
    lro = pd.read_csv(os.path.join(DATA_DIR,'fxmulti_results/logret_gran_out.csv'), index_col=0, parse_dates=True)
    lro.index = lro.index.date
    sto = get_formatted_summary_stats_from_df(lro)
```

Stats for the combination strategies

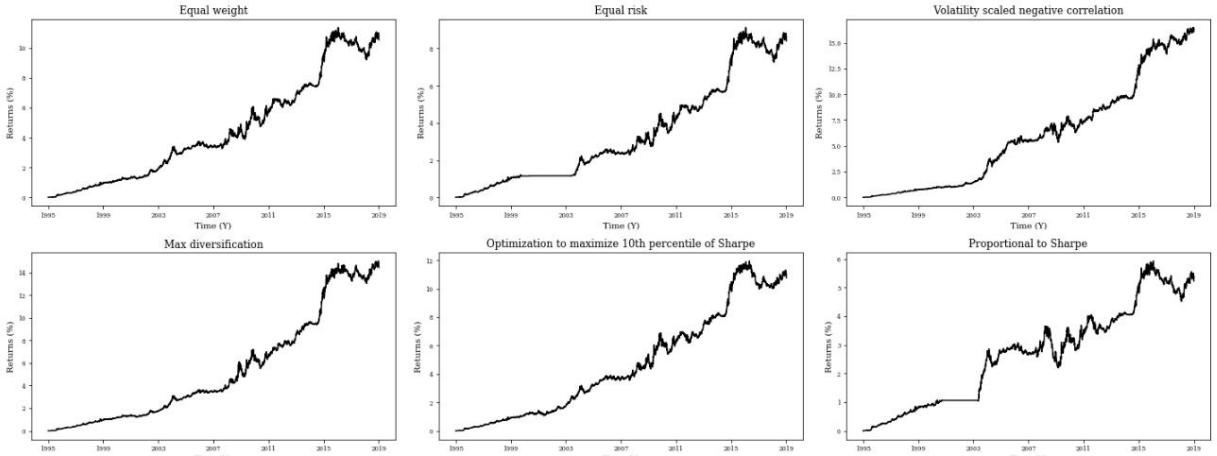
[39]:
```
sto.columns = [x.replace('_all_outsample','').replace('_', ' ').title() for x in sto.columns]
sto = sto.rename({'Invcorrelation By Sigma': 'Volatility scaled negative correlation',
                  'Inverse Correlation Dot Sharpe':'Negative correlation dot Sharpe',
                 'Max Rolling 10 Pctile Sharpe':'Optimization to maximize 10th percentile of Sharpe',
                 'Optimize Sharpe':'Optimization to maximize Sharpe',
                 'Max Diversification':'Max diversification',
                 'Equal Risk':'Equal risk',
                 'Equal Weight':'Equal weight',
                 'Proportional To Sharpe':'Proportional to Sharpe',
                 'Equal Risk Contribution':'Equal risk contribution'}, axis=1)
sto.loc[['annualized_returns', 'annual_sharpe_ratio_daily_returns'], :].T.sort_values('annual_sharpe_ratio_daily_returns', ascending=False)
```

[39]:

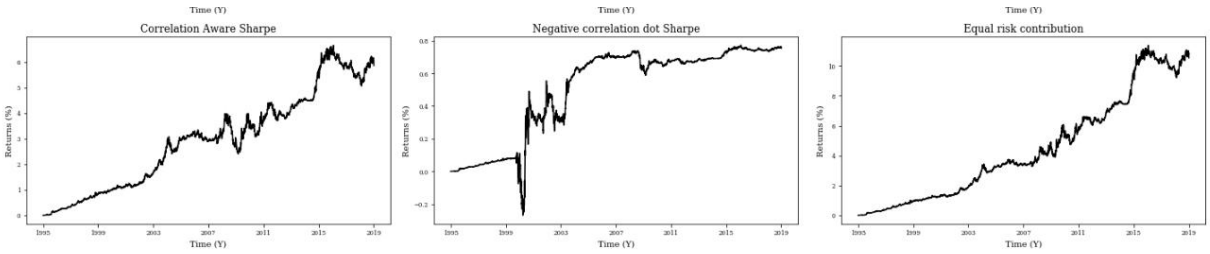
stat	annualized_returns	annual_sharpe_ratio_daily_returns
Volatility scaled negative correlation	10.73	0.88
Max diversification	10.33	0.84
Optimization to maximize 10th percentile of Sharpe	9.26	0.74
Equal risk contribution	9.19	0.73
Equal weight	9.19	0.73
Equal risk	8.39	0.65
Correlation Aware Sharpe	7.16	0.53
Proportional to Sharpe	6.79	0.49
Optimization to maximize Sharpe	6.15	0.43
Negative correlation dot Sharpe	2.04	0.03

Charts for the combination strategies

[40]:
```
import matplotlib.pyplot as plt
from stratdev.performance.performance_utils import *
plt.rc('font', family='serif')
plt.rc('xtick', labelsize='x-small')
plt.rc('ytick', labelsize='x-small')
lr = lro.copy()
lr.columns = [c.replace('_all_outsample','').replace('_', ' ').title() for c in lr.columns]
lr = lr.rename({'Invcorrelation By Sigma': 'Volatility scaled negative correlation',
                  'Inverse Correlation Dot Sharpe':'Negative correlation dot Sharpe',
                 'Max Rolling 10 Pctile Sharpe':'Optimization to maximize 10th percentile of Sharpe',
                 'Optimize Sharpe':'Optimization to maximize Sharpe',
                 'Max Diversification':'Max diversification',
                 'Equal Risk':'Equal risk',
                 'Equal Weight':'Equal weight',
                 'Proportional To Sharpe':'Proportional to Sharpe',
                 'Equal Risk Contribution':'Equal risk contribution'}, axis=1)
fig = plt.figure(figsize=(20, 15))
for i in range(9):
    ax = fig.add_subplot(4, 3, i+1)
    x = lr[lr.columns[i]].cumsum().apply(log_to_nominal)
    ax.plot(x,  color='k', ls='solid')
    ax.set_xlabel('Time (Y)')
    ax.set_ylabel('Returns (%)')
    plt.title(lr.columns[i])#.replace('_all_outsample','').replace('_', ' ').title())
    plt.tight_layout()
plt.savefig(os.path.join(DATA_DIR,'fxmulti_results/combination_charts_2.png'))
plt.show()
```



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Conclusion

Selecting indicators per products and aggregating gives the best results

[ ]:

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